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Revolutionizing OBM Performance with Nano Latex Material Raising the Bar

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Abstract

Wellbore stability is a critical component of successful well execution, influenced by factors such as effective mechanical stabilization, minimal chemical interaction with the surrounding formation, and maintaining a consistent pressure environment around the wellbore. While Oil-Based Mud (OBM) generally provides strong performance in these areas, it can fall short in highly demanding applications. This study evaluates a liquid additive designed to enhance OBM's sealing capacity and overall wellbore stabilization. To evaluate the anticipated enhancement of OBM performance through the addition of nano-latex-based products, the project was divided into two stages: laboratory assessment and field deployment. The laboratory evaluation involved identifying the optimal formulations for improvement, ensuring OBM compatibility with the selected nano-latex additive, and assessing its impact on fluid loss and filter cake enhancement. The best technical fluid option was proposed and subsequently deployed in field applications, where performance data was gathered to assess the impact of these modifications using data analytics. The incorporation of the nano-latex molecule (NLM) via a liquid carrier into the fluid design significantly reduced spurt loss and total fluid loss volumes under both standard (500 psi) and high-pressure (2500 psi) conditions across various temperature environments. The assessment also demonstrated that the NLM is compatible with a wide range of non-aqueous fluid chemicals, offering flexibility in designing drilling fluid systems to meet diverse operational requirements without bringing any detrimental impact on overall fluid properties. Results show a reduction of High Temperature High Pressure (HTHP) fluid loss by approximately 40%, and a reduction in spurt and total permeability plugging apparatus fluid loss by over 30%. Field data corroborated these laboratory findings, showing fewer instances of wellbore instability, related operational issues and further benefits in terms of fluid management such as minimization of bridging package treatments for specific applications.

The incorporation of NLM in OBM formulations has significantly enhanced sealing capabilities and ensured higher wellbore stability. This approach is anticipated to undergo further validation for reservoir applications, focusing on potential formation production impairment and the efficiency of removal processes.

Introduction

One of the paramount factors on oil and gas industry is to deliver wells within a constrained budget envelope and timeframe that fits field development strategies. Thus, reproducible and reliable operation well construction execution represents a cornerstone piece. On a more granular scale, the selected drilling fluid could either expedite or jeopardize this overall goal. From an environmental and logistical point of view, water-based mud (WBM) tends to provide an appealing profile. However, WBM could fall short to address highly demanding operational and technical requirements. OBM tends to overcome most of those challenges, but at a higher environmental footprint, bigger dedicated logistical requirements, and potential higher overall costs.

While OBM generally provides strong performance, it can fall short in highly demanding applications with unstable wellbores. Wellbore stability is a critical component of successful well execution, influenced by factors such as effective mechanical stabilization, minimal chemical interaction with the surrounding formation, and maintaining a consistent pressure environment around the wellbore. Extensive studies have been done and evolved on fluid bridging capabilities (Vickers et al., 2007). However, OBM faces limitations when it has to create an effective and resilient seal in tight formations or wide range of permeabilities in the same open hole trajectory, either due to unexpected formation changes, solid load limitations or particle selection.

Optimization in WBMs have shown that effective multi-layer approaches can deliver remarkable sealing capability along with chemical inhibition in front of highly unstable shale formations (Mehtar et al., 2010). Thus, WBM can potentially replace OBM up to a certain point, as demonstrated by Saragi et al. 2024); however, it might not be sufficient on more demanding performance requirements.

Capitalizing on these knowledge and experience, OBM is prone to further enhancement while incorporating non-traditional sealants such as modified latex or NLM. This specialty products work synergistically with OBM fluid loss additives and bridging packages, and provides pressure isolation due to sealing capacity at nano-size levels as demonstrated in the region by Mehtar and Saragi's work. Also, shale gas and oil drilling have increased the amount drilled footage throughout "shale" formations increasing the need to have relevant solutions to improve reliability and performance in extended reach applications. Under these scenarios reproducibility and scale volume tend to represent deciding variables for solution selection and deployment.

From a broader perspective, O&G industry efforts valued any option to reduce carbon footprint aligned, with its long-term goals. OBM treated with nano-sealants could provide a potential tangible impact on minimizing wellbore stability events, thus reducing operational time and the associated resource consumption in a shorter timeframe, and consequently the associated carbon footprint.

This study covers a technical assessment and field evaluation of nano-sealants or NLM usage with OBM demonstrating its flexibility and suitability for solving complex operational demands.

Laboratory Evaluation

The laboratory development of the Nano-Latex Molecule (NLM) started focusing on aqueous fluids and targeted specifically the enhancement of the filtrate control under harsh conditions while being compatible with a variety of different chemicals. Surely the benefits would have to be accompanied by a minimum variation in overall fluids properties (pH, rheology, etc…). This was successfully achieved as per demonstrated by Mehtar, 2010 and Friedheim, 2012 among others.

The anecdotal data indicates that NLM enable an enhancement of the sealing capacity of the existing package of bridging solids present in the drilling fluid beyond the concept described in the ideal packing theory.

The performance of NLM indicated that its capacity to control fluid loss could aid in stabilizing shales through the control of microfractures. The concept is that an existing microfracture in a given shale would have its aperture sealed with a much greater efficiency with the presence of NLM incorporated in the drilling fluid.

Conventional tool kit for fluid performance assessment do not offer the capability to assess microfracture plugging and pressure transmission control capabilities. Thus, the team considered to assess available information through a fit to purpose test: Shale Membrane Test.

The Figure 1 shows the schematic of the Shale Membrane Test (SMT). The equipment consists of a chamber with an upper portion and a lower portion separated by a shale sample with an existing microfracture. The SMT test consist of Injecting the testing fluid using the injection pump through aperture P2 using a 200 psi to 300 psi pressure.

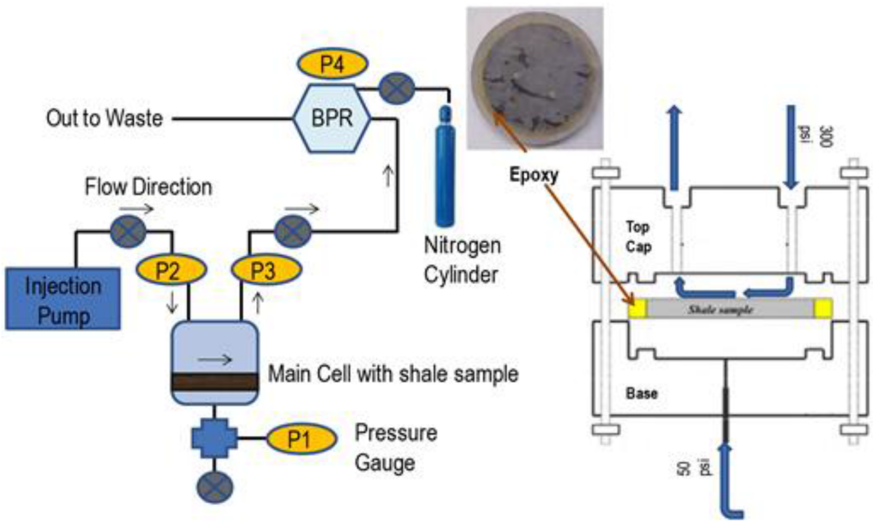

Figure 1—Shale Membrane Test Schematic (Friedheim, 2012)

The tested fluid gets in contact with the previously prepared shale core sample (Figure 2) where the microfracture is exposed to the fluid at a given pressure and temperature and the fluid leaves the cell through aperture P3. The shale sample (Figure 2) is monitored through the aperture P1 where a pressure of 50 psi is applied and monitored since the test starts. The pressure variation in P1 is the indication that the microfracture propagated from one side of the shale to the other.

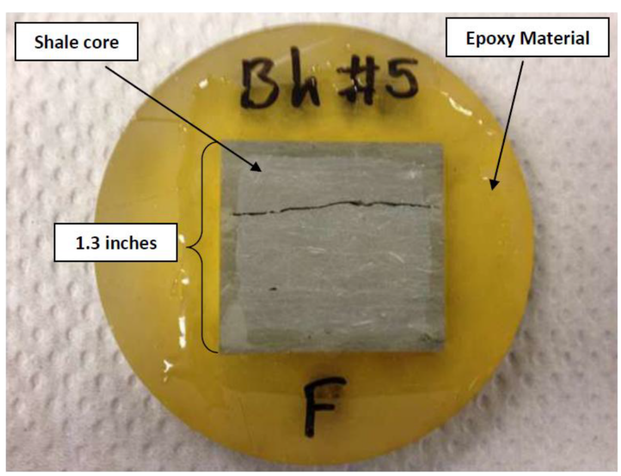

Figure 2—Shale Core Sample with existing microfracture

The samples used the same fluid for all tests summarized in Figure 3 except for the NLM that was added in some of the tests and was absent in others.

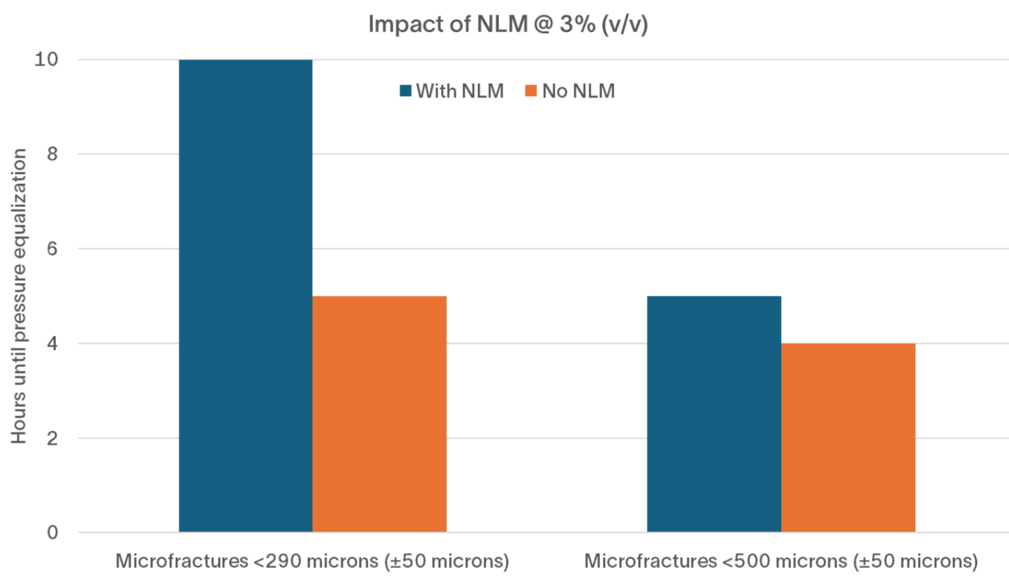

Figure 3—Summary of SMT tests

The smaller fractures tend to have a substantial enhancement of their stability having the fracture propagation time being doubled with the presence of NLM (increment of 100% of the time as seen above). Larger fractures tend to have an enhancement of stabilization in the range of 25% time increment, but given the nano size nature of the NLM its benefits are not so pronounced.

The NLM material was tested with non-aqueous fluid to demonstrate its performance in tightening the fluid loss using a variety of chemicals and diverse base fluids as follows.

The formulations presented in Table 1 presented a remarkable similar rheological profile

Table 1—Low Temperature Non-Aqueous Fluid

Product	F1 (ppb)	F2 (ppb)
Diesel	195	187.82
Lime	6	6
Wetting Agent	3	3
Drill Water	57	55
CaCl2	20.36	19.61
Organophilic Clay	6	6
Nano-Latex Molecule	n/a	10.5
Asphaltic Resin	6	6
Emulsifier	8	8
Barite	140.71	139.15

The rheological profile has minimum variation due the presence of NLM and the fluid loss is reduced substantially (reduction of 63%). Following, the same fluid was used for testing on a permeability plugging apparatus (PPA) test, and the results demonstrated the improvement after the NLM addition.

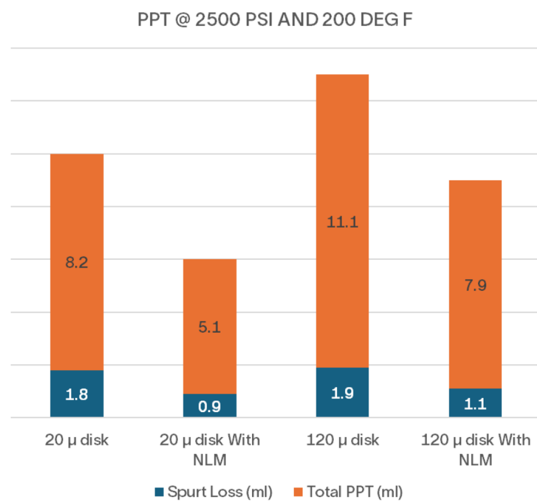

Figure 4—Summary of NLM performance in PPA test

Again, the Figure 4 demonstrates the presence of NLM substantially reduces the overall fluid low without any additional chemical and with no adverse effects on rheology as per Table 2. It is critical to assess the Spurt Loss, one of the key drivers to prevent differential stuck pipe, is reduced by 50% in the 20 microns disk and by 42% in the 120 microns disk demonstrating that the NLM is effective in a wide range of apertures. However, again the performance of the NLM is better in smaller diameter apertures as previously showed.

Table 2—Summary of properties from Low Temperature Non-Aqueous Fluid

Mud Weight (ppg)	10.5	10.5
Rheology Temp. (F)	150	150
600 rpm	53	59
300 rpm	35	39
200 rpm	26	31
100 rpm	19	22
6 rpm	9	10
3 rpm	8	9
Gels 10"	9	11
Gels 10’	10	13
LSYP	7	8
HTHP FL (ml)	7	2.6
HTHP Pressure (psi)	500	500
HTHP Temp. (F)	250	250
Cake (mm)	2	1

The addition of NLM in recycled non-aqueous fluids was also assessed to understand the impact of adding it on a fluid containing low gravity solids while also changing the initial oil to water ratio (O/W) to a different value. As per Table 3 – Impact of NLM in Recycled Non-Aqueous fluid, the addition of the NLM at 1.5% and 3% volume by volume (v/v) changes, as expected, the O/W ratio with minor impact in the overall rheological profile of the fluid. It is interesting to highlight that the 3% v/v addition drives to a substantial 50% reduction of the fluid loss on a fluid that already had a very tight fluid loss from a starting point.

Table 3—Impact of NLM in Recycled Non-Aqueous fluid

		Initial	1.5%NLM (v/v)	3% NLM (v/v)
Rheology:	Temp. ºF	150	150	150
600 rpm		80	106	109
300 rpm		49	66	69
200 rpm		37	51	54
100 rpm		26	35	38
6 rpm		11	14	14
3 rpm		12	13	13
Gels 10″	lbs/100ft2	12	14	14
Gels 10′	lbs/100ft2	14	17	17
Plastic Visc.	cP	31	40	40
Yield point	lbs/100ft2	18	26	29
LSYP	lbs/100ft2	13	12	12
Media	µ	Paper	Paper	Paper
Temperature	ºF	250	250	250
Differential pressure	psi	500	500	500
HTHP fluid loss	ml	1.2	1	0.6
Water	%	10	11	12.5
Oil	%	60	59	59.1
Uncorrected solid	%	30	30	29.9
Oil / Water Ratio		86/14	84/16	83/17

The change in O/W ratio can be a concern in several different operational scenarios thus the assessment of the addition of NLM in a recycled non-aqueous fluid was performed as per Figure 5 where it is clear that the variation of the O/W given the optimum addition of 3% v/v of NLM has marginal impact in the coefficient of friction of the fluid.

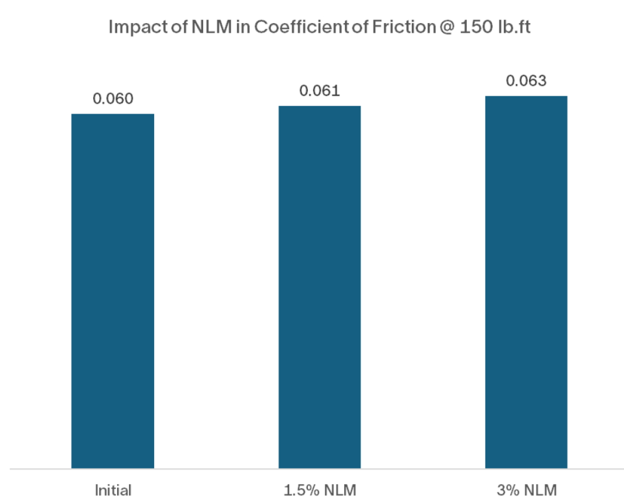

Figure 5—Impact of NLM in Coefficient of Friction

Field Cases

Field laboratory evaluations demonstrated the merits and potential benefits of NLM in OBM formulations, thus, the drilling team decided to move on to deploy this modification in multiple demanding scenarios.

Case #1: Reducing Bridging Material in Deep, High-Temperature Wells

Challenging Well Conditions and Bridging Material Limitations. In a deep drilling operation targeting a lateral section of up to 6,000 ft at a total depth of 21,000 ft, the well faced extreme conditions: temperatures ranging from 280° F to 310° F, and overbalance pressures as high as 2,500 psi. Traditional bridging material treatments, while effective, posed significant associated challenges. These included increased low gravity solids (LGS%) and difficulties in removing ground-up bridging particles during drilling, which exacerbated drill string friction and required frequent dilution to control solids levels. The environmental and operational drawbacks of excessive bridging material use — such as high-volume generation and carbon footprint — also demanded a more sustainable solution.

Innovative Solution: NANO LATEX MOLECULE as a Game-Changer. The introduction of NLM into the oil-based mud system revolutionized the approach. By reducing bridging material additions by up to 80%, the NLM effectively controlled fluid loss while maintaining the critical permeability plugging test (PPT) within the upper limit of the targeted values. Despite reaching the target threshold, the benefits were undeniable: reduced solids content, improved mud recycling efficiency, and a significant drop in environmental impact. The polymer's ability to form a stable filter cake without compromising formation integrity allowed achieving well objectives while minimizing operational complexity and guaranteeing safe trouble-free operation.

Operational and Environmental Benefits. The shift to NLM resulted in measurable cost savings, with drilling fluid costs reduced by 8–13% across the entire well due to less bridging package chemical consumption. The need for dilution was also minimized, further lowering operational expenses. While drilling parameters like rate of penetration (ROP) and torque remained consistent across 20 wells, the reduction in solids content likely contributed to cleaner hole conditions and reduced waste. Perhaps most importantly, the carbon footprint was slashed by decreasing reliance on bridging materials, aligning with industry sustainability goals. Also, stuck pipe incidents remain to a minimum occurrence level. Overall fluid properties were managed within programmed ranges as shown on Table 4.

Table 4—OBM + NLM properties for High Temperature application

		40 ppb Bridging + 1% NLM	6 ppb Bridging + 2% NLM
READING	L600	56	57
READING	L300	35	35
READING	L200	22	24
READING	L100	14	16
READING	L6	8	14
READING	L3	7	8
GEL STRENGHT 10 SEC	Lb/100ft2	8	7
GEL STRENGHT 10 MIN	Lb/100ft2	11	9
PLASTIC VISCOSITY	CP	21	22
YIELD POINT	Lb/100ft2	14	13
HPHT TEMPERATURE	F	275	275
HPHT PRESSURE	Psi	500	500

		40 ppb Bridging + 1% NLM	6 ppb Bridging + 2% NLM
HPHT FLUID LOSS	cc/30 min	4	6
HPHT CAKE THICKNESS	1/32″	2	2
PPT DISC NUMBER	Micron	40	40
PPT TEMPERATURE TEST	F	275	275
PPT PRESSURE TEST	Psi	1500	1500
PPT TOTAL FLUID LOSS	cc/30 min	5.4	7.2
PPT SPURT LOSS	cc/30 sec	0.6	1
PPT CAKE THICKNESS	1/32″	2	2
MUD WEIGHT	**pcf**	**88**	**88**

The reduction of used Bridging material not only saved cost, but also provided more environmental and sustainable approach to control fluid loss as shown in below Table 5. The calculation method used for CO2 footprint reduction was detailed by Gilbert, 2010.

Table 5—Summary of CO2 transportation emission footprint reduction after applying NLM

# of wells	Average quantity of sacks per well	Predicted Transportation emissions per well	Total CO2 emission avoided
20	1148	1500 kg of CO2	60 Tons

Case #2: Overcoming Shale Instability in a Faulted Formation

Geological Challenges in a Complex Shale Formation. Drilling through a shale formation with weak bedding planes and a composition of 55–70% quartz, 25–35% clay minerals (kaolinite and illite), and carbonate-rich interlayers posed severe stability risks. The presence of multiple faults and the absence of prior drilling data in the area compounded the challenge. The shale's sensitivity to pressure fluctuations made it prone to collapse, with fluid loss invasion exacerbating lubrication between platelets —a phenomenon that could trigger cavings and pack-offs during pressure drops (Figure 6 and Figure 7).

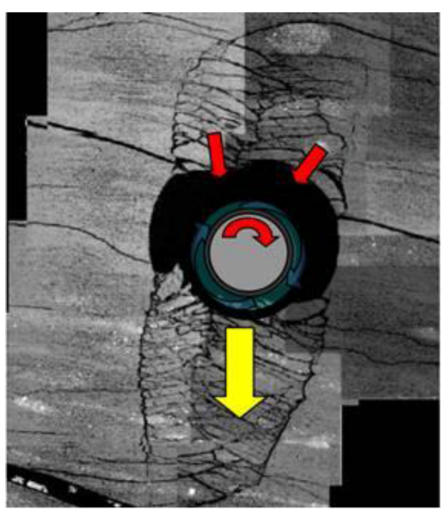

Figure 6—The effects of several faults in the reservoir while drilling and how it may impact the hole stability. From SLB Geomechanics Website

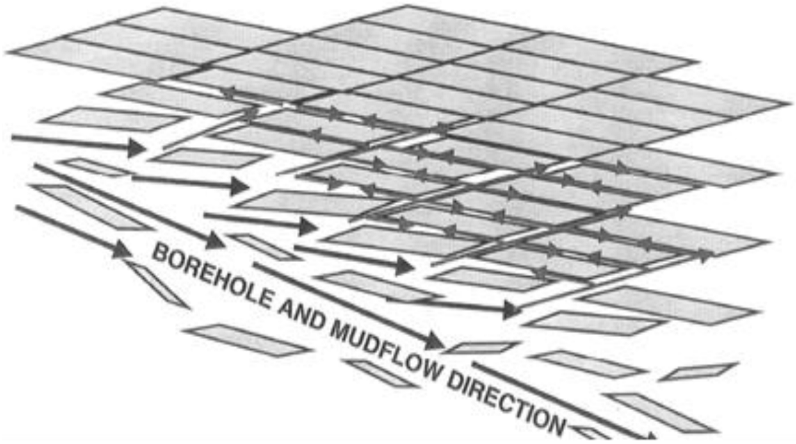

Figure 7—The forces of mud flow towards Weak Bedding Planes of the shale formation

Symptoms of Hole Instability and Operational Risks. The instability manifested as increased torque, pump pressure spikes, and excessive shale cavings on the shale shaker; see Figure 8. Tight spots and mud losses due to packing-off forced a sidetrack, highlighting the urgency of a solution. Without intervention, the risk of well abandonment or costly re-drilling loomed large.

Figure 8—samples of shale cavings that prevent the completion of first sidetrack which resulted in total hole collapse.

NLM's Role in Enhancing Shale Stability. By increasing the NLM dosage, the drilling team addressed the root cause of instability: pore pressure transmission and fluid invasion into microfractures. The NLM's ability to seal nano-pores and prevent fluid lubrication between shale platelets drastically reduced the likelihood of cavings and collapse; this translates in shorter operational time, see Table 6. This mechanism stabilized the formation, even under pressure fluctuations, by maintaining structural integrity and minimizing the destabilizing effects of fluid loss.

Table 6—Summary of sidetrack footage and elapsed time

Sidetrack	NLM (% v/v)	Days	From (ft)	To (ft)
#1	1	58 days	15651	19306
#2	3.5	33 days	15635	21000

Optimizing Drilling Fluid Properties for Stability. To further mitigate equivalent circulating density (ECD) effects, the drilling fluid's rheology was carefully adjusted while maintaining optimal density. This balance ensured that the NLM solution did not compromise pumpability or pressure management, critical for safe and efficient drilling.

Conclusions

The laboratory evaluations followed by field deployment of NLM treated OBM demonstrated its value as an option to enhance performance, and open a new set of opportunities for fluid design and applications. The following major points reflect laboratory and field application findings:

- Lab results show a reduction of HTHP fluid loss by approximately 40%, and a reduction in spurt and total permeability plugging apparatus fluid loss by over 30%.
- No significant impairment on fluid loss rheology or lubricity after adding 3 % NLM on conventional OBM formulations
- In case study #1, a field multi-well campaigned recorded savings up to 80 % on bridging package consumption
- In case study #2, the NLM successfully helped stabilize the unstable shale formation, preventing cavings and collapse, ensuring safe, cost-effective well completion in a complex, and faulted reservoir.
- In case study #2, NLM increase on concentration from 1 to 3.5 % v/v allowed a faster operation with 25 days savings

Further evaluations could consider formation damage assessment to validate NLM treated OBM applications in sections contemplating open hole completions, or depleted reservoirs limited on density requirements. Also, NLM could be considered as a major pillar on OBM design aiming to replace conventional components such as organophilic clays and treated particulates for fluid loss control.

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